Section 3.1 IM-OLS Cointegration Estimation Matrix: U.S. NFC Transformation Elasticity (1929–2024)

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Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.1 of Chapter 2. All models are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over the full NIPA sample window (1931–2024, $T = 94$ observations after listwise deletion for the lag-1 conditioning variables). To resolve multi-collinearity, all interactions are orthogonalized via Frisch-Waugh-Lovell (FWL) against their base regressors and any detected Bai-Perron step dummies.

1 Specification Set 1: Double Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1}$)

Table 1: Specification Set 1: Double Interaction FWL-Orthogonalized IM-OLS

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\alpha}$ (Intercept)	6.48	8.57	6.31	8.54	6.51	9.03	6.34	8.76	7.16	6.57	7.00	6.27
	(0.28)	(0.70)	(0.28)	(0.67)	(0.27)	(0.68)	(0.27)	(0.68)	(0.41)	(1.00)	(0.39)	(0.96)
$\hat{\beta}$ (Private Core k_{priv})	0.59	0.37	—	—	0.58	0.33	—	—	0.52	0.57	—	—
	(0.02)	(0.06)			(0.02)	(0.06)			(0.04)	(0.09)		
$\hat{\beta}$ (Infra-Augmented k_{aug})	—	—	0.59	0.37	—	—	0.58	0.35	—	—	0.52	0.58
			(0.02)	(0.06)			(0.02)	(0.06)			(0.03)	(0.08)
$\hat{\phi}$ (ω_{t-1})	0.14	-4.03	0.66	-3.43	-1.09	-3.71	-0.58	-3.08	-3.14	-2.92	-3.05	-2.84
	(2.11)	(1.55)	(2.06)	(1.42)	(1.82)	(1.55)	(1.76)	(1.45)	(1.34)	(1.42)	(1.25)	(1.32)
$\hat{\theta}$ (Orthogonalized Interaction)	0.22	2.38	0.38	2.27	1.24	1.56	1.39	1.69	0.22	1.93	0.48	2.10
	(1.41)	(0.75)	(1.34)	(0.70)	(1.01)	(0.70)	(0.95)	(0.66)	(0.51)	(0.56)	(0.48)	(0.51)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	1.30	1.30	1.31	1.31	1.42	1.42	1.44	1.44	4.53	4.53	4.57	4.57
VIF (ω)	1.30	1.30	1.31	1.31	1.42	1.42	1.44	1.44	4.53	4.53	4.57	4.57
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.84	-3.69	-2.75	-3.71	-2.91	-3.76	-2.82	-3.66	-3.20	-3.41	-3.07	-3.40
Shin LM (C_{Shin}) [H_0: Coint.]	2.90***	0.83***	3.03***	0.72***	3.35***	0.80***	3.49***	0.76***	1.01***	0.21***	1.10***	0.23***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	5940.48***	7099.79***	6782.26***	6281.43***	8526.80***	7995.14***	9232.10***	6492.54***	1531.29***	1169.61***	848.02***	1688.86***
Bandwidth Ratio (b)	0.2095	0.1587	0.2127	0.1375	0.1959	0.1601	0.1980	0.1387	0.2596	0.2079	0.2532	0.1975
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Stars for coefficient rows and the Residual VW Type 2 cointegration test are derived from a custom Monte Carlo simulation suite mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, applying FWL orthogonalization in simulation). Distributional variables centered at 1929 initial state (t_0).

2 Specification Set 2: Triple Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1} \cdot (1 - \iota_{t-1})$)

Table 2: Specification Set 2: Triple Interaction FWL-Orthogonalized IM-OLS

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\alpha}$ (Intercept)	7.51	8.57	7.36	8.41	7.57	8.67	7.40	8.46	8.83	8.83	8.71	8.71
	(0.35)	(0.18)	(0.37)	(0.18)	(0.30)	(0.17)	(0.33)	(0.17)	(0.36)	(0.36)	(0.35)	(0.35)
$\hat{\beta}$ (Private Core k_{priv})	0.49	0.37	—	—	0.48	0.36	—	—	0.37	0.37	—	—
	(0.03)	(0.02)			(0.03)	(0.01)			(0.03)	(0.03)		
$\hat{\beta}$ (Infra-Augmented k_{aug})	—	—	0.49	0.38	—	—	0.48	0.37	—	—	0.37	0.37
			(0.03)	(0.02)			(0.03)	(0.01)			(0.03)	(0.03)
$\hat{\phi}$ (ω_{t-1})	5.17	8.40	5.57	8.01	5.88	8.41	6.14	8.44	13.71	13.71	13.42	13.42
	(2.57)	(1.53)	(2.65)	(1.48)	(2.23)	(1.55)	(2.30)	(1.48)	(3.37)	(3.37)	(3.20)	(3.20)
$\hat{\gamma}$ ($\omega_{t-1} \cdot (1 - \iota_{t-1})$)	-7.67	-11.29	-7.54	-10.44	-8.28	-11.24	-8.07	-10.78	-17.63	-17.63	-17.35	-17.35
	(2.26)	(1.42)	(2.36)	(1.39)	(1.99)	(1.42)	(2.09)	(1.37)	(3.40)	(3.40)	(3.24)	(3.24)
$\hat{\theta}$ (Orthogonalized Interaction)	3.14	1.68	3.08	1.60	2.88	1.52	2.84	1.48	0.78	0.78	0.89	0.89
	(0.40)	(0.17)	(0.40)	(0.16)	(0.31)	(0.15)	(0.32)	(0.14)	(0.21)	(0.21)	(0.19)	(0.19)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	No	No	No
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	13.10	13.10	13.71	13.71	13.31	13.31	13.97	13.97	16.79	16.79	17.79	17.79
VIF (ω)	15.64	15.64	15.76	15.76	17.76	17.76	17.90	17.90	114.80	114.80	119.39	119.39
VIF (x_3)	39.60	39.60	40.93	40.93	44.40	44.40	45.88	45.88	188.43	188.43	197.89	197.89
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-3.17	-5.56***	-3.95	-5.50***	-4.28*	-5.52***	-4.12	-5.58***	-4.32*	-4.32*	-4.38*	-4.38*
Shin LM (C_{Shin}) [H_0: Coint.]	1.50***	0.07	1.65***	0.08	1.64***	0.09	1.70***	0.09	0.34***	0.34***	0.35***	0.35***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	663.23***	31878.27***	3431.25***	106077.72***	714.64***	38696.66***	3986.02***	130150.47***	66416.80***	66416.80***	82435.27***	82435.27***
Bandwidth Ratio (b)	0.1672	0.1779	0.1389	0.1477	0.1709	0.1782	0.1407	0.1480	0.1724	0.1724	0.1484	0.1484
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Stars for coefficient rows and the Residual VW Type 2 cointegration test are derived from a custom Monte Carlo simulation suite mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, applying FWL orthogonalization in simulation). Distributional variables centered at 1929 initial state (t_0).